UNDER REVIEW

RESCIENCE C

Replication / Geosciences

[Re] An open and extensible framework for spatially explicit land use change modelling: the lulcc R package

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1 Introduction

Moulds, Buytaert, and Mijic[1] presented lulcc, an R package for spatially explicit land use change modelling, with an explicit aim: to make land use change models open, so that their results can be reproduced and their code examined and extended. They demonstrated the package on the Plum Island Ecosystems data set, which ships with it, going through the stages of an inductive model: suitability from logistic regression, demand from interpolation between observed maps, spatial allocation by CLUE-S[2] and by an ordered procedure[3], and pattern validation by three-map comparison[4].
We replicate that application with a different implementation in a different language. The replication runs on DisSModel[5], a Python framework for spatially explicit dynamic models, and its land use package DisSLUCC[6]. The CLUE-S allocation of DisSLUCC was ported from LuccME, the land use extension of TerraME[7] developed at INPE, and was written independently of lulcc; both lulcc and LuccME implement the algorithm from its published description[2]. The comparison therefore tests two independent readings of the same description, and lulcc's own demonstration, which was written to be reproducible, provides the reference.

2 The original model

The Plum Island Ecosystems data set[8] covers a site in north-east Massachusetts on a grid of 434×497 cells of about 100 m (99.92 m by 99.95 m), 113 563 of them inside the study area. It holds land use maps for 1985, 1991 and 1999 (forest, built, other), treated as time points 0, 6 and 14, and three explanatory variables: elevation, slope and distance to built land in 1985.

Suitability. One binary logistic regression per class is fitted on a stratified random sample of 10% of the cells of the 1985 map: built on the three variables, forest and other on elevation and slope. The paper also fits regression trees and random forests and compares the three model types by the area under the ROC curve (AUC) on the remaining cells (its Fig. 5), and evaluates the glm for built on the cells that became built between 1985 and 1991 (its Fig. 6).

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The authors have declared that no competing interests exist.
Code is available at https://github.com/LambdaGeo/rescience-lulcc..
Data is available at https://github.com/LambdaGeo/lulcc-goldens.

Demand. The number of cells of each class at every time point from 0 to 14 is interpolated linearly between the three observed maps and rounded to integers that sum to the number of cells.

CLUE-S allocation. At each time step every cell receives, for each class, its suitability plus an elasticity (0.2 for all classes) if the cell already belongs to that class. Each cell is allocated to the class with the highest value plus a class-specific iteration term; the term is then increased by the difference between demand and allocated area, times a scale factor (10^{-6}), until the largest difference is at most 50 cells. At every iteration the suitability is perturbed by a uniform random number in $[-0.0002, 0.0002]$ (`jitter.f`).

Ordered allocation. Classes are visited in a fixed order (built, forest, other). A class with rising demand takes the most suitable cells not yet settled; a class with falling demand releases its least suitable cells. In the stochastic version a cell qualifies only if its suitability exceeds (or, when releasing, falls below) a uniform random number.

Validation. The initial observed map (1985), the final observed map (1999) and the simulated map for 1999 are compared at eight resolutions, aggregating 2×2 to 256×256 cells, through the three-dimensional contingency table of Pontius, Peethambaram, and Castella[4]. The paper reports the agreement budget and the figure of merit of the forest-to-built transition for both allocation methods (its Figs. 7 and 8).

3 Methods

3.1 Reference

The paper sets no random seed, and three of its steps draw random numbers: the training sample, the CLUE-S perturbation and the ordered allocation. The numbers it prints are one draw from a distribution and cannot be repeated exactly, not even with lulcc. We therefore generated the reference ourselves by running the paper's demonstration script (`demo/gmd-paper.R`) with lulcc 1.0.4[9] in a fixed environment, and archived the outputs[10]. Two kinds of reference were produced:

- a *deterministic* one, with the randomness removed (seed 1 for the training sample, `jitter.f` $= 0$, ordered allocation not stochastic), exported stage by stage: training cells, regression coefficients, the suitability matrix, the demand table, all simulated maps and the three-map tables;
- an *ensemble* of 30 runs of the script as published, seeds 1 to 30.

3.2 Implementation

The replication reads the data set from GeoTIFF files exported from the R package and uses no R code. Suitability is fitted with statsmodels; the AUC is computed by the Mann–Whitney statistic, with ties counted as one half. CLUE-S is the allocation component of DisSLUCC version 0.5.0 (`AllocationDClueSLike`), used unchanged, on a DisSModel raster backend; the environment starts at time 1, as lulcc does. Three pieces that DisSLUCC lacks were written for this replication: the random perturbation of CLUE-S (a subclass of the DisSLUCC component), the ordered allocation, and the three-map comparison with its agreement budget and figure of merit.
The paper describes these last pieces at the level of the algorithm. Where it leaves a choice open we read lulcc's source code and followed it, so that differences in the comparison would point at the algorithm rather than at a convention. The points resolved this way are:

- demand rounding: rows rescaled to the number of cells, floored, and the remaining cells given to the classes with the largest remainders;
- training sample: the class codes are numeric, so the stratification of `caret::createDataPartition`, which lulcc calls, groups them by quantile, which for this map gives two strata, {forest, built} and {other};
- ordered allocation: which cells are candidates, the temporary class for released cells, the limit of 1000 random draws, and ties kept in cell order;
- three-map comparison: the layout of the table, aggregation blocks anchored at the upper-left corner with incomplete edge blocks weighted by their number of valid cells, and empty denominators left out of the sums.

The CLUE-S allocation itself was not modified to match lulcc.

3.3 Comparison and acceptance criteria

Criteria were recorded in the code repository before either ensemble was run (the commit precedes those with results). At the deterministic level, fed with the reference's own inputs, each stage must agree within numerical precision and the allocated maps must be identical. At the ensemble level, our 30 runs (seeds 1 to 30 of NumPy's default generator) are compared with lulcc's 30 on 20 metrics: the AUC of the glm for each class and for the gain of built, and the figure of merit of forest to built for each allocation method at each of the eight resolutions. A metric passes if the means differ by at most three standard errors of the difference and a two-sample Kolmogorov–Smirnov test gives $p \geq 0.01$. The regression trees and random forests of the paper's Fig. 5 are outside the criteria: we did not reimplement R's rpart and randomForest, and other implementations would differ by construction.

4 Results

4.1 Deterministic level

Table 1 lists the stages. The CLUE-S allocation of DisSLUCC, given lulcc's suitability and demand, produced the same map as lulcc at each of the 14 time steps, cell by cell; it needed 51 iterations at the first step and 257 at the last. The ordered allocation also matched at every step, and the three-map tables and figures of merit agreed to rounding error. Running the whole chain from our own regression gave the same final map: the coefficients differ by about 10^{-9} in relative terms, which did not change any allocation.

Stage	Result
Observed transitions 1985–1999 (paper, Sect. 3.2)	identical
Regression coefficients, same training cells	max. relative difference 1.3×10^{-9}
Suitability from lulcc's coefficients	max. difference 9×10^{-14}
Suitability from our coefficients	max. difference 2.8×10^{-10}
Demand, $t = 0, \ldots, 14$	identical
CLUE-S, 14 steps	0 cells differ at every step
Ordered allocation, 14 steps	0 cells differ at every step
Three-map tables, 8 resolutions, both methods	max. difference 1.1×10^{-16}
Figure of merit forest → built	max. difference 2.2×10^{-16}
Whole chain from our own regression	final map identical

Table 1. Deterministic level: each stage fed with the reference's inputs.

4.2 Ensemble level

All 20 metrics passed. Table 2 shows a selection and Figure 1 the figure of merit at every resolution; the spread between runs is the same in both implementations, including the larger spread of the stochastic ordered allocation.

Metric	lulcc mean (sd)	replication mean (sd)	diff.	tol.	KS p
AUC glm, forest	0.6439 (0.0007)	0.6437 (0.0006)	−0.0003	0.0005	0.07
AUC glm, built	0.9382 (0.0003)	0.9382 (0.0003)	−0.0001	0.0002	0.81
AUC glm, other	0.7295 (0.0010)	0.7290 (0.0008)	−0.0004	0.0007	0.14
AUC glm, gain of built	0.6446 (0.0004)	0.6446 (0.0004)	0.0000	0.0003	0.81
FoM CLUE-S, factor 2	0.1168 (0.0017)	0.1167 (0.0019)	−0.0001	0.0014	0.81
FoM CLUE-S, factor 16	0.3766 (0.0058)	0.3756 (0.0062)	−0.0010	0.0046	0.81
FoM CLUE-S, factor 256	0.7684 (0.0052)	0.7688 (0.0052)	+0.0004	0.0040	0.96
FoM ordered, factor 2	0.1079 (0.0017)	0.1079 (0.0019)	0.0000	0.0014	1.00
FoM ordered, factor 16	0.4311 (0.0188)	0.4308 (0.0199)	−0.0003	0.0150	0.96
FoM ordered, factor 256	0.6795 (0.0326)	0.6784 (0.0327)	−0.0011	0.0253	1.00

Table 2. Ensemble level, 30 runs each (selection of the 20 metrics; all passed). FoM: figure of merit of the forest-to-built transition at 1999. Tolerance: three standard errors of the difference of means.

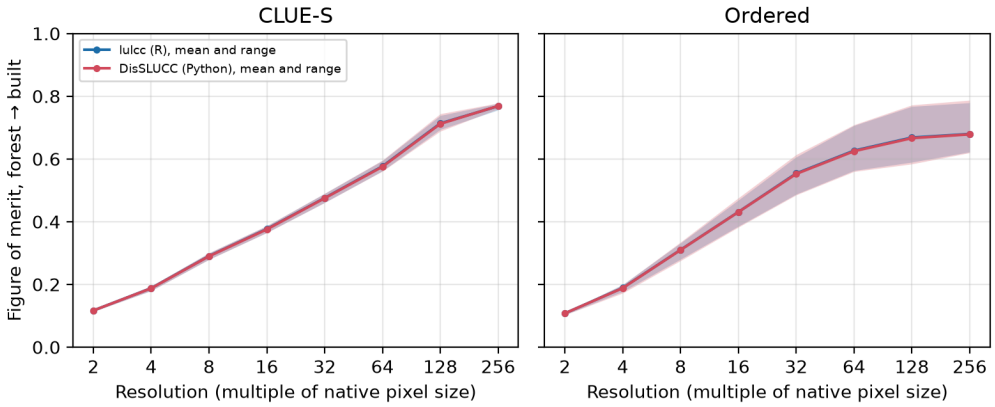


Figure 1. Figure of merit of the forest-to-built transition against resolution, as in Fig. 8 of the original article: mean and range of 30 runs of lulcc and of the replication.

The values printed in the paper fall inside lulcc's own spread with one exception. In our 30 runs of lulcc, the paper's AUC for built (0.9384) is at the 70th percentile, that for the gain of built (0.6448) at the 63rd and that for other (0.7276) at the 7th; the AUC for forest (0.6423) is slightly below the lowest of the 30 runs (0.6425). The paper used lulcc 1.0 with the R of 2015; our reference used lulcc 1.0.4 on current R. We did not investigate the difference further.

5 Obstacles

The demonstration script does not run unchanged. With current versions of the raster and sp packages, the CLUE-S and ordered allocations stop with `different crs`. The rasters of the data set were serialised by an old version of raster, before the slot that holds the coordinate reference system in WKT form existed; the allocation creates each simulated map from the initial one, the new object carries a CRS in the current form and the old one does not, and the check that precedes stacking them fails. Reassigning the

CRS of the old objects does not help: it leaves them with no CRS at all. Rebuilding each layer as a new raster object with the same grid, values and CRS solves it; this is the only change we made to the script. A package written to make land use models reproducible thus needed a repair to run its own published example eleven years later, for a reason unrelated to the model.

An undeclared dependency. lulcc loads the `gsubfn` package when it builds the explanatory variables, but lists it only as a suggestion; it has to be installed by hand.

No seed. Without a seed, the paper's numbers can only be placed within a distribution, which is why the reference had to be generated and why the comparison has an ensemble level.

Details visible only in the source code. Besides the points listed in the methods, two behaviours of lulcc are worth noting. The random forest models receive a numeric 0/1 response and therefore run in regression mode. The CLUE-S loop has a second stopping condition based on the mean signed difference between demand and allocation; because the rounded demand sums to the number of cells, that mean is zero and only the largest difference stops the iteration.

6 Conclusion

The Plum Island Ecosystems application of lulcc was replicated. With the same inputs, an independent implementation of CLUE-S reproduces lulcc's maps exactly, and so do our ordered allocation and validation; with independent random numbers, the two implementations agree in distribution on every metric compared. The main obstacle was not in the model but in the environment around it.

Reproducing this replication

Code: https://github.com/LambdaGeo/rescience-lulcc. Reference outputs and the Docker environment that produces them: https://github.com/LambdaGeo/lulcc-goldens. With the reference downloaded, `make env`, `make check` (about one minute), `make ensemble` (about 15 minutes) and `make compare` reproduce Tables 1 and 2 and Figure 1. The software versions are fixed in the repository.

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Complete table of the 20 ensemble metrics: see `results/ensemble_comparison.csv` *in the code repository.*

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